

DATA STRUCTURES - Examination 2020

Total Marks: 70

SECTION A

1.

Attempt all questions in brief.

- (a) Define stack and queue.
- (b) What is a threaded binary tree?
- (c) State the time complexity of selection sort.
- (d) What is a sparse matrix?
- (e) Define a binary search tree.
- (f) What is a graph? Define its types.
- (g) What is a hash function?

SECTION B

2.

Attempt any three of the following:

- (a) Explain push and pop operations of a stack with an algorithm.
- (b) Explain polynomial addition using linked lists.
- (c) Explain quick sort and derive its average case complexity.
- (d) Explain Warshall's algorithm for shortest paths.
- (e) Explain hashing and different collision handling methods.

SECTION C

3.

Attempt any one part of the following:

- (a) Explain binary search tree with insertion and deletion examples.
- (b) Explain Dijkstra shortest path algorithm with an example.

4.

Attempt any one part of the following:

- (a) Write the binary search algorithm and explain its complexity.
- (b) Explain AVL tree rotations with examples.